

Particle Filter for Real-time Estimation and Compensation of Nonlinear Friction

Yoshihiko Suzuki¹, Jun'ya Fukui², Gan Chen³ and Isao Takami³

Abstract—The friction deteriorates the performance of the positioning control. The friction characteristics depend on operating conditions such as the normal force, the ambient temperature and so on, which vary with time. A friction estimation method is proposed to compensate in real-time. The friction has nonlinear characteristics including the static friction, the Coulomb friction, the Stribeck effect and the hysteresis effect. The characteristics are described by the nonlinear dynamical model called the LuGre model. The nonlinear friction is estimated in real-time by the particle filter. This filter is suitable for the estimation of the friction because it can be applied to the nonlinear and non-Gaussian state space model. The problem of the particle filter is the calculation cost. The calculation cost is decreased and the estimation accuracy is increased by countermeasures which are the selection of the state variables and the resampling algorithm. The effectiveness of the proposed method is illustrated by the experiment through the comparison with the unscented Kalman filter and evaluated by root mean square error. The error without friction compensation is 101.4×10^{-7} while the one with friction compensation using the unscented Kalman filter is 15.4×10^{-7} and with friction compensation using the particle filter is 8.7×10^{-7} .

I. INTRODUCTION

The friction is one of main factors deteriorating the performance of the positioning control which is installed in robots, machine tools, semiconductor manufacturing equipment and medical equipment. The friction causes some undesirable phenomena, e.g. the stationary error, the tracking delay, the stick slip motion, the limit cycle, and quadrant glitches. To decrease the impact of the friction, the ball screw system is applied to the positioning control system. However, that is not enough for precise control. The simple error feedback controller can not prevent the stationary error and the tracking delay. It is necessary to design the control system which proactively compensates the friction. In this study, friction model based compensator is designed. The friction characteristics depend on time-varying conditions such as the position, the normal force, the ambient temperature, the lubricant condition, and the material wear [1]–[4]. Therefore, it is required to estimate and compensate the friction in real-time.

Many models are proposed to describe the friction characteristics. The Coulomb model and the Tustin model represent static characteristics of the friction [5]. The Dahl model expresses the dynamics of the friction [6]. The LuGre model

is the dynamical one to describe the transition from the pre-sliding regime to the sliding regime [7]. On the LuGre model, the contact surface is considered as the aggregation of bristles, and friction characteristics are approximated by the elasticity and viscosity of bristles. The Leuven model is an improving one of the LuGre model to describe the hysteresis characteristics of the friction [8]. The generalized Maxwell-slip model is a friction model which connects elements of the friction in parallel to express the hysteresis characteristics [9].

The friction has nonlinear characteristics, e.g. the static friction, the Coulomb friction, the Stribeck effect, and the hysteresis effect. The nonlinear filters such as the extended Kalman filter (EKF), the unscented Kalman filter (UKF) and the particle filter (PF) are applied to the estimation of the nonlinear friction [10]. The UKF is the method which estimates the state of nonlinear systems by the approximation of probability density function at sampling points called sigma points [11]. The PF is the state estimation algorithm which approximates the probability distribution by Monte Carlo simulation using particles [12]–[14]. The EKF and the UKF assume that the system noise, the observation noise, and the estimated state value are Gaussian, while the PF does not require this assumption and it is applied to very wide classes [15]. In comparison with the EKF and the UKF, the PF accurately estimates nonlinear systems. The PF is applied to the state estimation for systems with the hysteresis [16], the target tracking [17], the localization for robots [18] and the fault diagnosis [19]. Several methods are proposed in the study of the friction estimation using the PF [20] and [21].

The papers of the friction compensation have been published, e.g. the disturbance observer (DOB) [22], the observer based on the LuGre model [23], the UKF based on the LuGre model [24] and the estimating parameters of the friction [25] are proposed. The methods which identify parameters of the friction beforehand and compensate by the feedforward control [26] and [27]. The adaptive control using the LuGre model are described in [3] and [28].

In this study, the performance of positioning control on a ball screw is improved by the real-time estimation and compensation of uncertain nonlinear friction which depends on operating conditions. The LuGre model is applied to estimate the nonlinear friction because it describes static characteristics including the static friction, the Coulomb friction and the Stribeck effect, and the dynamic characteristic which is the transition from the pre-sliding regime to the sliding regime. As a result, quadrant glitch is predicted

¹Yoshihiko Suzuki is with Graduate School of Science and Engineering, Nanzan University, pikurusu-50.824@ezweb.ne.jp

²Jun'ya Fukui is with Mitsubishi Electric Corporation

³Gan Chen and Isao Takami are with Faculty of Science and Engineering, Nanzan University

when the friction direction is inverted. Using the model, the nonlinear friction is represented by less parameters. The PF is applied to estimate varying friction characteristics which have strong nonlinearity. The accuracy of the estimation is improved by using the PF which is appropriate for the estimation of nonlinear and non-Gaussian systems because it does not assume that systems are linear or Gaussian. One of the problems of the PF is the calculation cost. The reduction of the calculation cost is needed to apply it in real-time. To improve the accuracy of the estimation and the efficiency of the calculation, the PF is carried out with some countermeasures, e.g. the selection of state variables and the resampling algorithm. The contribution of this study is to propose the method to estimate and compensate the nonlinear friction in real-time by combining the PF with the LuGre model. The effectiveness of the proposed method is illustrated by the experiment through the comparison with the UKF.

II. MODELING

In this section, the model of the ball screw system for the precise positioning control is described and the friction model is introduced.

A. Ball Screw System

The ball screw system is shown in Fig. 1. It consists of two axes, A and B. Each axis is composed of a ball screw, a table, a guide rail and a motor. The ball screw converts the rotating motion of the motor to the linear motion of the table. The axis B is on the axis A. The vertical load of the axis A is different from the axis B.

The motion equations of the table and the motor are given as follows:

$$M\ddot{x}_p = K(L\theta - x_p) - (\sigma_2\dot{x}_p + F_{nl}), \quad (1)$$

$$J_\theta\ddot{\theta} = T - LK(L\theta - x_p), \quad (2)$$

where x_p is the position of the table, θ is the angle of the motor, T is the torque of the motor, M is the mass

of the table, J_θ is the moment of inertia of the rotating system, σ_2 is the viscous friction coefficient, K is the spring coefficient of the ball screw, F_{nl} is the nonlinear friction, and L is the ball screw lead, which is the coefficient converting the rotational motion of the motor to the linear motion of the table. The ball screw is stiff, and the strain of axis is negligible. Therefore, $L\theta = x_p$ can be assumed. Then using (1) and (2), the dynamical model of the ball screw system is obtained as follows:

$$(J_\theta + L^2M)\ddot{\theta} = -L(\sigma_2L\dot{\theta} + F_{nl}) + T. \quad (3)$$

The state space representation of the ball screw system is obtained as follows:

$$\begin{cases} \dot{x} &= Ax + Bu - BLF_{nl}, \\ y &= Cx, \end{cases} \quad (4)$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{L^2\sigma_2}{J_\theta + L^2M} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{1}{J_\theta + L^2M} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

where $x = [\theta \ \dot{\theta}]^T$ is the state vector, $u = T$ is the input, $y = \theta$ is the output.

B. Servo System

The servo system is designed for the state space representation (4). Let $x_e = [\theta \ \dot{\theta} \ \eta]^T$ be the state vector where $\eta = \int(\theta_{ref} - \theta)dt$ is the integral of the error between the reference θ_{ref} and the output $y = \theta$. Then, the augmented system is obtained as follows:

$$\begin{cases} \dot{x}_e &= A_e x_e + B_e u, \\ y &= C_e x_e, \end{cases} \quad (5)$$

$$A_e = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}, \quad B_e = \begin{bmatrix} B \\ 0 \end{bmatrix}, \quad C_e = \begin{bmatrix} C & 0 \end{bmatrix},$$

The augmented system (5) is discretized with the sampling time T_s . The discrete model is obtained as follows:

$$\begin{aligned} x_{ed}(k+1) &= A_{ed}x_{ed}(k) + B_{ed}u(k), \\ A_{ed} &= e^{A_e T_s}, \quad B_{ed} = \int_0^{T_s} e^{A_e \tau} B_e d\tau, \end{aligned} \quad (6)$$

where the state vector is $x_{ed}(k) = [\theta_d(k) \ \dot{\theta}_d(k) \ \eta_d(k)]^T$. The state feedback gain is designed by the discrete-time linear quadratic regulator (DLQ).

C. Nonlinear Friction

Several kinds of model are proposed to describe the nonlinear friction characteristics. The higher accuracy of the friction model, the more parameters are required. Among friction models, the LuGre model is chosen for this study. Using the LuGre model, the nonlinear friction characteristics are represented by less parameters. In the LuGre model, the contact surface is considered as the aggregation of bristles. The friction characteristics are approximated by the elasticity and the viscosity of bristles. The mathematical expression of the LuGre model is described as follows:

$$F = \sigma_0 z + \sigma_1 \dot{z} + \sigma_2 L \dot{\theta}, \quad (7)$$

$$\dot{z} = L \dot{\theta} - \sigma_0 \frac{L|\dot{\theta}|}{g(\dot{\theta})} z, \quad (8)$$

$$g(\dot{\theta}) = (F_c + (F_s - F_c)\exp(-L|\dot{\theta}|/v_s)), \quad (9)$$

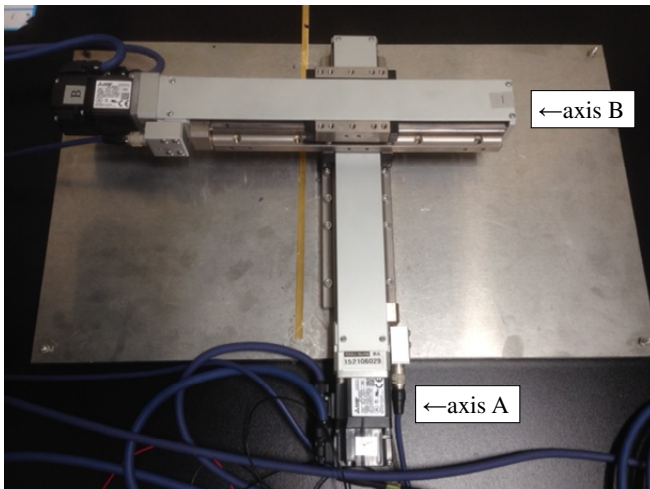


Fig. 1. Ball screw system

where z is the position of the bristle, σ_0 is the spring coefficient of the bristle, σ_1 is the damper coefficient of the bristle, F_c is the Coulomb friction, F_s is the static friction, and v_s is the Stribeck velocity. The first and second terms of (7) represent the nonlinear friction F_{nl} . The third term of (7) represents the viscous friction. The equation (8) represents the dynamics of the bristle. The function $g(\theta)$ represents the static characteristic of the friction.

III. FRICTION ESTIMATION USING PF

In the previous section, nonlinear friction characteristics is represented by the LuGre model. However, this characteristics depend on the operating conditions, e.g. the position, the normal force, the ambient temperature, the lubricant condition and the material wear. Since the friction characteristics vary with time, it is necessary to estimate in real-time for the precise positioning control. It is known that the PF accurately estimates states and parameters in the system with the hysteresis [16]. The estimation error of the UKF does not converge to zero, while the estimation error of the PF converges to zero as the number of particles approaches infinity [21]. In this section, the method is proposed to estimate and compensate the nonlinear friction using the PF in real-time.

A. Particle Filter

The discrete-time probabilistic model is given as follows:

$$\begin{cases} x(k+1) = f_k(x(k)) + w(k) \\ y(k) = h_k(x(k)) + v(k), \end{cases} \quad (10)$$

where $x(k)$ is the state vector, $y(k)$ is the observation vector, $w(k) \sim N(0, Q)$ is the system noise, $v(k) \sim N(0, R)$ is the observation noise, f_k and h_k are the nonlinear functions.

The minimum variance estimation of $x(k)$ is the conditional expectation which can be described as follows:

$$\hat{x}(k) = E\{x(k)|Y^k\} = \int x(k)p(x(k)|Y^k)dx(k), \quad (11)$$

where $\hat{x}(k)$ is the estimated value of $x(k)$, $Y^k = \{y(0), \dots, y(k)\}$ represents the set of the observations ($0 \leq t \leq k$), and $p(x(k)|Y^k)$ is the conditional probability density function of $x(k)$ given Y^k . The probability density function $p(x(k)|Y^k)$ is represented by (12) and (13),

$$p(x(k)|Y^k) = \frac{p(y(k)|x(k))p(x(k)|Y^{k-1})}{p(y(k)|Y^{k-1})}, \quad (12)$$

$$p(x(k+1)|Y^k) = \int p(x(k+1)|x(k))p(x(k)|Y^k)dx(k). \quad (13)$$

The conditional probability distributions (12) and (13) are called the filter distribution and the predictive distribution, respectively. The state vector is estimated by using the PF with (12) and (13). The equations (12) and (13) are approximated by the empirical distribution functions as follows:

$$p(x(k)|Y^k) \simeq 1/n \sum_{i=1}^n \delta(x(k) - x^{(i)}(k|k)), \quad (14)$$

$$p(x(k)|Y^{k-1}) \simeq 1/n \sum_{i=1}^n \delta(x(k) - x^{(i)}(k|k-1)), \quad (15)$$

where $\delta(x)$ is the Dirac delta function, n is the number of particles, $x^{(i)}(k|k)$ is the filtered particle, and $x^{(i)}(k|k-1)$ is the one step predicted particle for $i = 1, \dots, n$. Using (14) and (15), particles and these weights approximate the distribution of the state vector.

The algorithm of the PF is described as follows:

- 1) Generate the initial particles $x^{(i)}(0|0)$ for $i = 1, \dots, n$ and define $k = 1$.
- 2) Prediction step (for $i = 1, \dots, n$)
 - Generate samples of the system noise $w^{(i)}(k-1)$ by using normal random numbers.
 - Calculate the predicted state value $x^{(i)}(k|k-1) = f_k(x^{(i)}(k-1|k-1), u(k-1)) + w^{(i)}(k-1)$.
- 3) Filtering step (for $i = 1, \dots, n$)
 - Calculate weights of particles $\alpha^{(i)}(k) = \exp\left(-\frac{1}{2}\{\bar{y}^{(i)}(k)\}^T R^{-1}\bar{y}^{(i)}(k)\right)$, where $\bar{y}^{(i)}(k) = y(k) - h_k(x^{(i)}(k|k-1))$.
 - Normalize weights by $\tilde{\alpha}^{(i)}(k) = \alpha^{(i)}(k) / \sum_{i=1}^n \alpha^{(i)}(k)$.
- 4) Resampling step (for $i = 1, \dots, n$)
 - Find j which satisfies the conditions: $\sum_{l=1}^{j-1} \alpha^{(l)}(k) < \xi^{(i)} \leq \sum_{l=1}^j \alpha^{(l)}(k)$, where $\xi^{(i)} = \frac{i-0.5}{n}$. This is the deterministic algorithm [13].
 - Set $x^{(i)}(k|k) = x^{(j)}(k|k-1)$.
- 5) Calculate the estimated value $\hat{x}(k) = 1/n \sum_{i=1}^n x^{(i)}(k|k)$.
- 6) $k \leftarrow k+1$ and back to Prediction step.

Using the above algorithm, the nonlinear friction is estimated.

B. Equation for Friction Estimation

In this section, nonlinear dynamical equations are derived to estimate the friction. By discretizing the dynamics of bristles (8), the following equation is obtained:

$$\begin{aligned} z_d(k+1) &= \exp\left(-\frac{\sigma_0 L |\dot{\theta}_d(k)|}{g(\dot{\theta}_d(k))} T_s\right) z_d(k) \\ &+ \left\{ 1 - \exp\left(-\frac{\sigma_0 L |\dot{\theta}_d(k)|}{g(\dot{\theta}_d(k))} T_s\right) \right\} \frac{g(\dot{\theta}_d(k))}{\sigma_0} \text{sgn}(\dot{\theta}_d(k)) \\ &= f_z(\dot{\theta}_d(k), z_d(k), g(\dot{\theta}_d(k))). \end{aligned} \quad (16)$$

It is necessary for the friction estimation to identify three parameters F_s , F_c and v_s . On the estimation by the PF, the number of particles increases according to the number of estimated variables. The PF has the trade-off relation between the calculation cost and the estimation accuracy. Using the PF for real-time estimation, the number of variables has to be reduced. In this paper, $g(\dot{\theta})$ is estimated representatively instead of estimating three parameters respectively. In pre-sliding regime, which is the time period from stop to start moving, the parameter F_s is dominant over the function $g(\dot{\theta})$. The function $g(\dot{\theta})$ is regarded as an unknown parameter g_p . The parameter g_p is assumed to be constant, i.e. $\frac{d}{dt}g_p = 0$.

Using (4) and (16), the discrete-time nonlinear representation is obtained as follows:

$$\begin{aligned}\hat{x}_L(k+1) &= f_L(\hat{x}_L(k), u(k)) + \tilde{w}(k), \\ \hat{y}_L(k) &= h_L(\hat{x}_L(k)) + \tilde{v}(k), \\ f_L(\hat{x}_L(k), u(k)) &= \begin{bmatrix} A_d \begin{bmatrix} \hat{x}_1(k) \\ \hat{x}_2(k) \end{bmatrix} + B_d(u(k) - L\hat{F}_{nl}(k)) \\ f_z(\hat{x}_2(k), \hat{x}_3(k), \hat{x}_4(k)) \\ \hat{x}_4(k) \end{bmatrix}, \\ h_L(\hat{x}_L(k)) &= \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \hat{x}_L(k), \\ A_d &= e^{AT_s}, B_d = \int_0^{T_s} e^{A\tau} B d\tau, \\ \hat{F}_{nl}(k) &= \sigma_1 L \hat{x}_2(k) + \sigma_0 (1 - \sigma_1 \frac{L|\hat{x}_2(k)|}{\hat{x}_4(k)}) \hat{x}_3(k), \end{aligned} \quad (17)$$

where $\hat{x}_L(k) = [\hat{x}_1(k) \ \hat{x}_2(k) \ \hat{x}_3(k) \ \hat{x}_4(k)]^T = [\hat{\theta}_d(k) \ \hat{\theta}_d(k) \ \hat{z}_d(k) \ \hat{g}_{pd}(k)]^T$ is the state vector, $\hat{y}_L(k)$ is the output, $u(k)$ is the input, $\tilde{w}(k) \sim N(0, \hat{Q})$ is the system noise, and $\tilde{v}(k) \sim N(0, \hat{R})$ is the observation noise. Using (17), $\hat{x}_L(k)$ and $\hat{F}_{nl}(k)$ are estimated.

The control system combining the DLQ and the friction compensator using the PF is shown in Fig. 2. The variables $\hat{F}_{nl}$ and $\hat{\theta}_d$ are estimated by the PF. The estimation of the nonlinear friction $\hat{F}_{nl}$ is applied to the input and the effect of the nonlinear friction F_{nl} is canceled. The DLQ is applied to the positioning control using the observation $y = \theta$ and the estimation $\hat{\theta}_d$. First, the influence of friction is removed by the compensator. If error still occur, the DLQ controller will remove it. The controller has integrator and brings the stationary error to zero.

C. Real-time Estimation using the PF

In this study, the friction characteristics are estimated in real-time. In the case of the high accuracy estimation using the PF, numerous particles are required and the calculation cost increases. Then, the following countermeasures are applied to reduce the calculation cost of the PF without loss of the estimation accuracy.

- Estimated parameters in the friction dynamics are replaced by only one parameter g_p instead of three parameters F_s , F_c and v_s .

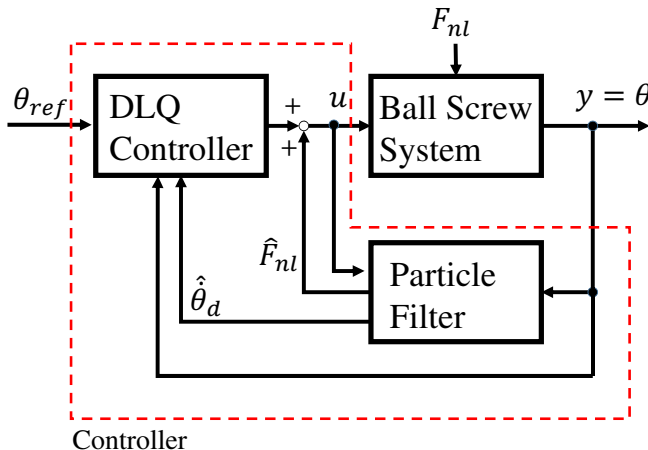


Fig. 2. Diagram of control system

- In the prediction step of the PF, the calculation time is needed to generate the system noise using normal random numbers. The random numbers are generated and stored in the memory beforehand.
- The deterministic algorithm is chosen for the resampling. The deterministic algorithm makes the calculation error more accurate and the calculation cost more efficient than the roulette wheel selection. In the case of the roulette wheel selection, random numbers are generated as many as the number of particles. Then, it costs calculation time and the results vary. The deterministic algorithm does not need to generate random numbers, and it is the efficient algorithm in terms of calculation cost.

The attention should be paid to how the system noise is given. The covariance matrix $\hat{Q}$ is tuned in the experience. When the estimation value diverges or changes suddenly step-like, the degeneracy (impoverishment), which is decreasing the diversity of particles, occurs. In this case, it is needed to increase the variance of the system noise. When the value of estimation is oscillated, the accuracy of the estimation is increased by decreasing the values of diagonal components of $\hat{Q}$. Thus, the best variance of system noise is the smallest one which does not cause the degeneracy.

IV. EXPERIMENT

In this section, the effectiveness of the proposed method is illustrated by the experimental results. The impact of the friction appears most largely at the reversal action of the table. To show the impact of the friction remarkably, the circular trajectory is traced using the two axes driven by the ball screw system. The reference of the axis A is given as the sinusoidal signal, and that of the axis B is given as the cosinusoidal signal. The amplitude of the reference is 10^{-4} m, and the frequency of the reference is 1/8Hz. The variation of the table position is so small and so slow to estimate the nonlinear friction in pre-sliding regime. The sampling time T_s is set to 10^{-3} s. The encoder resolution is about 7.6nm. Three methods are executed for the comparison. The first method is the proposed one shown in Fig. 2. The number of particles is 800 on the PF. This is described as “DLQ and PF” in this section. The second is the one applying the UKF instead of the PF in Fig. 2. This is described as “DLQ and UKF”. The third is the one applying the DLQ only which does not include the friction estimation and compensation. This is described as “DLQ” only.

The experimental results are shown in Fig. 3. The horizontal axis is the table position of the axis A, and the vertical axis is the table position of the axis B. The fine dotted line is the reference, the solid line shows the trajectory of “DLQ and PF”, the dashed line shows the trajectory of “DLQ and UKF”, and the dashed and dotted line shows the trajectory of “DLQ”. By the experimental results in Fig. 3, when applying “DLQ” only, quadrant glitches are observed. When applying “DLQ and UKF”, quadrant glitches are disappeared, while the error is observed. Using “DLQ and PF”, quadrant glitches are not observed and the trajectory draws the almost

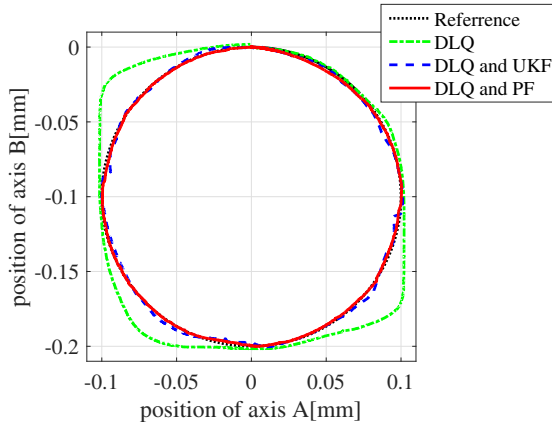


Fig. 3. Experimental results

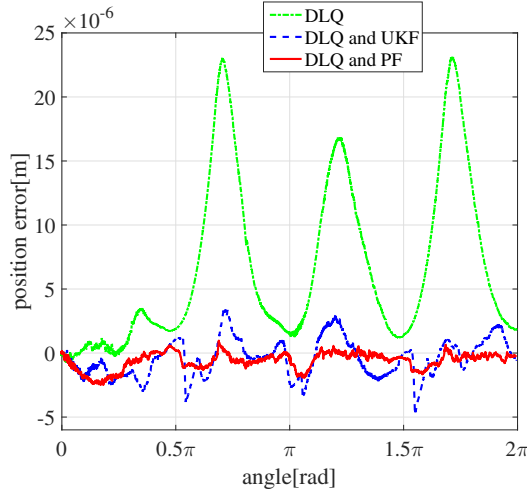


Fig. 4. Tracking error in experiment

complete circle. The tracking errors between the reference and the trajectory in the experiment are shown in Fig. 4. The horizontal axis is angle, and the vertical axis is the tracking error. The tracking error is evaluated by root mean square error (RMSE). RMSE of “DLQ” is 101.4×10^{-7} while RMSE of “DLQ and UKF” is 15.4×10^{-7} and RMSE of “DLQ and PF” is 8.7×10^{-7} . The tracking error of “DLQ and PF” is the smallest of them and the tracking performance is improved by the proposed method. In the experiment, the estimation of nonlinear friction $\hat{F}_{nl}$ is shown in Fig. 5. The horizontal axis is time, and the vertical axis is the estimation of nonlinear friction of the axis A. The solid line shows the estimated nonlinear friction by the PF, and the dashed line shows the one by the UKF. The difference in the nonlinear friction $\hat{F}_{nl}$ between using the UKF and the PF is especially observed when the friction direction is inverted. Using the UKF, the overshoot and the oscillation phenomenon observed in the estimated nonlinear friction $\hat{F}_{nl}$. By using $\hat{F}_{nl}$ of the UKF, too large input is applied to the ball screw system. As a result, they cause the trajectory deviation from the reference. On the contrary, using the PF, the overshoot of $\hat{F}_{nl}$ is not observed.

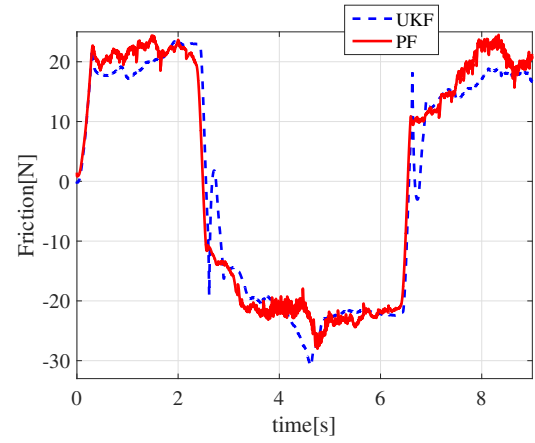


Fig. 5. Estimated friction $\hat{F}_{nl}$ in experiment

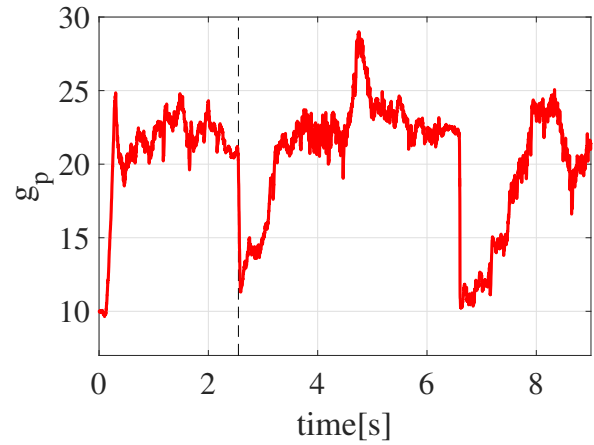


Fig. 6. Time response of $\hat{g}_p$

The non-Gaussianity of the PF is explained based on experimental results. The variation of the estimated value of $\hat{g}_{pd}$ with time is shown in Fig. 6. The value varies rapidly from 22N to 12N at 2.55s (dashed line). The histograms of $\hat{g}_{pd}$ of the PF at 2.45s, 2.55s and 2.59s, which is the diagram right before, during and right after it varies, are shown in Fig. 7 (a), (b) and (c), respectively. The unimodal distribution with a peak centered at 22N is shown in Fig. 7 (a). The non-Gaussian distribution with two peaks centered at 22N and 12N is shown in Fig. 7 (b). The unimodal distribution with a peak centered at 12N is shown in Fig. 7 (c). Fig. 7 (a), (b) and (c) represent the transition of the particles. This means the information is transmitted continuously by the particles of the PF, and the non-Gaussian distributions of the particles express the variation of the nonlinear friction characteristics properly. The PF is suitable to estimate the friction.

V. CONCLUSIONS

In this paper, the friction estimation method is proposed to compensate the friction in real-time. The friction has strongly nonlinear characteristics. They are described by the nonlinear dynamical model, the LuGre model. By using this

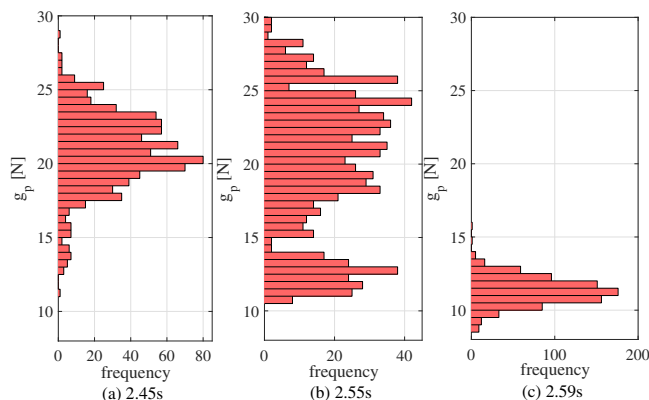


Fig. 7. Transition of particle distribution of $\hat{g}_p$. (a) at 2.45s and (b) at 2.55s and (c) at 2.59s.

model, the nonlinear friction is represented by less parameters. The nonlinear friction is estimated in real-time by the PF. This filter is suitable for the estimation of the friction because it can be applied to the nonlinear and non-Gaussian state space model. To increase the accuracy of estimation and the efficiency of calculation, some countermeasures are adopted to the PF. The effectiveness of the proposed method is illustrated by the experiment through the comparison with the UKF.

REFERENCES

- [1] Pai-Yi Huang, Yung-Yaw Chen, and Min-Shin Chen, "Position-dependent Friction Compensation for Ballscrew Tables", *IEEE International Conference on Control Applications*, pp. 863-867, 1998
- [2] C. L. Chen, M.J. Jang, and K.C.Lin, "Modeling and high-precision control of a ball-screw-driven stage", *Precision Engineering*, Vol. 28, No. 4, pp. 483-495, 2004
- [3] C. Canudas de Wit and P. Lischinsky, "Adaptive Friction Compensation with Partially Known Dynamic Friction Model", *International Journal of Adaptive Control and Signal Processing*, Vol. 11, No. 1, pp. 65-80, 1997
- [4] Wenjie Chen, Kyoungchul Kong, and Masayoshi Tomizuka, "Dual-Stage Adaptive Friction Compensation for Precise Load Side Position Tracking of Indirect Drive Mechanisms", *IEEE Transactions on Control Systems Technology*, Vol. 23, No. 1, pp. 164-175, 2015
- [5] Brian Armstrong-Hélouvry, Pierre Dupont, and C. Canudas de Wit, "A Survey of Models, Analysis Tools and Compensation Methods for the Control of Machines with Friction", *Automatica*, Vol. 30, No. 7, pp. 1083-1138, 1994
- [6] P. R. Dahl, "A Solid Friction Model", *The Aerospace Corporation*, TOR-0158(3107-18)-1, 1968
- [7] C. Canudas de Wit, H. Olsson, K. J. Åström, and P. Lischinsky, "A New Model for Control of Systems with Friction", *IEEE Transactions on Automatic Control*, Vol. 40, No. 3, pp. 419-425, 1995
- [8] J. Swevers, F. Al-Bender, Chris G. Ganseman, and T. Prajogo, "An Integrated Friction Model Structure with Improved Presliding Behavior for Accurate Friction Compensation", *IEEE Transactions on Automatic Control*, Vol. 45, No. 4, pp. 675-686, 2000
- [9] Farid Al-Bender, Vincent Lampaert, and Jan Swevers, "The Generalized Maxwell-Slip Model: A Novel Model for Friction Simulation and Compensation", *IEEE Transactions on Automatic Control*, Vol. 50, No.11, pp.1883-1887, 2005
- [10] G. G. Rigatos, "Nonlinear Kalman Filters and Particle Filters for integrated navigation of unmanned aerial vehicles", *Robotics and Autonomous Systems*, Vol. 60, No. 7, pp. 978-995, 2012
- [11] S. J. Julier and J. K. Uhlmann, "A New Extension of the Kalman Filter to Nonlinear Systems", *Proceedings of SPIE, Signal Processing, Sensor Fusion and Target Recognition VI*, Vol. 3068, pp. 182-193, 1997
- [12] G. Kitagawa, "A Monte Carlo Filtering and Smoothing Method for Non-Gaussian Nonlinear State Space Models", *Proceedings of the 2nd U.S.-Japan Joint Seminar on Statistical Time Series Analysis*, pp. 110-131, 1993
- [13] G. Kitagawa, "Monte Carlo Filter and Smoother for Non-Gaussian Nonlinear State Space Models", *Journal of Computational and Graphical Statistics*, Vol. 5, No. 1, pp. 1-25, 1996
- [14] N. J. Gordon, D. J. Salmond, and A. F. M. Smith, "Novel approach to nonlinear/non-Gaussian Bayesian state estimation", *IEEE Proceedings-F*, Vol. 140, No.2, pp. 107-113, 1993
- [15] K. Nakamura, T. Higuchi, and N.Hirose, "Application of Particle Filter to Identification of Tsunami Simulation Model", *Proceedings of Joint 3rd International Conference on Soft Computing and Intelligent Systems and 7th International Symposium on advanced Intelligent Systems*, pp. 1890-1895, 2006
- [16] Shahram Tafazoli, Clarence W. de Silva, and Peter D. Lawrence, "Tracking Control of an Electrohydraulic Manipulator in the Presence of Friction", *IEEE Transactions on Control Systems Technology*, Vol. 6, No. 3, pp. 401-411, 1998
- [17] Mark Pupilli and Andrew Calway, "Real-Time Camera Tracking Using a Particle Filter", *British Machine Vision Conference*, 2005
- [18] Sebastian Thrun, Wolfram Burgard, and Dieter Fox, "Probabilistic Robotics", *The MIT Press*, 2005
- [19] Daisuke Yasufuku, Akira Fujii, and Yutaka Yamamoto, "A New Approach to Fault Diagnosis of Rotating Machine by Using Particle Filter as Parameter Estimation", *Transactions of the Institute of Systems, Control and Information Engineers*, Vol. 25, No. 9, pp. 221-229, 2012
- [20] L. Zhang and J.C. Trinkle, "The Application of Particle Filtering to Grasping Acquisition with Visual Occlusion and Tactile Sensing", *IEEE International Conference on Robotics and Automation*, pp. 3805-3812, 2012
- [21] Nevena Stevanović, Peter L Green, Keith Worden, and Poul Henning Kirkegaard, "Friction estimation in wind turbine blade bearings", *Structural Control and Health Monitoring*, Vol. 23, No. 1, pp. 103-122, 2016
- [22] H. M. Kim, S. H. Park, J. H. Song, and J. S. Kim, "Robust position control of electro-hydraulic actuator systems using the adaptive backstepping control scheme", *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*, Vol. 224, No. 6, pp. 737-746, 2010
- [23] L. Freidovich, A. Robertsson, A. Shiriaev, and R. Johansson, "LuGre-Model-Based Friction Compensation", *IEEE Transactions on Control Systems Technology*, Vol. 18, No. 1, pp. 194-200, 2010
- [24] Jun'ya Fukui, Takayuki Yamamoto, Gan Chen, and Isao Takami, "Real-time Identification and Compensation of Asymmetric Friction Using Unscented Kalman Filter", *IEEE Conference on Control Technology and Applications*, pp. 1085-1090, 2017
- [25] A. Amthor, St. Zschäck, and Ch. Ament, "Position Control on Nanometer Scale based on an Adaptive Friction Compensation Scheme", *34th Annual Conference of IEEE Industrial Electronics*, pp. 2568-2573, 2008
- [26] Z. Jamaludin, H. V. Brussel, and J. Swevers, "Friction Compensation of an XY Feed Table Using Friction-Model-Based Feedforward and an Inverse-Model-Based Disturbance Observer", *IEEE Transactions on Industrial Electronics*, Vol. 56, No. 10, pp.3848-3853, 2009
- [27] N. A. Rafan, Z. Jamaludin, T. H. Chiew, L. Abdullah, and M. N. Maslan, "Contour error analysis of precise positioning for ball screw driven stage using friction model feedforward", *12th Global Conference on Sustainable Manufacturing*, Vol. 26, pp.712-717, 2015
- [28] E. Panteley, R Ortega, and M. Gäfvert, "An Adaptive Friction Compensator for Global Tracking in Robot Manipulators", *Systems & Control Letters*, Vol. 33, No. 5, pp. 307-313, 1998